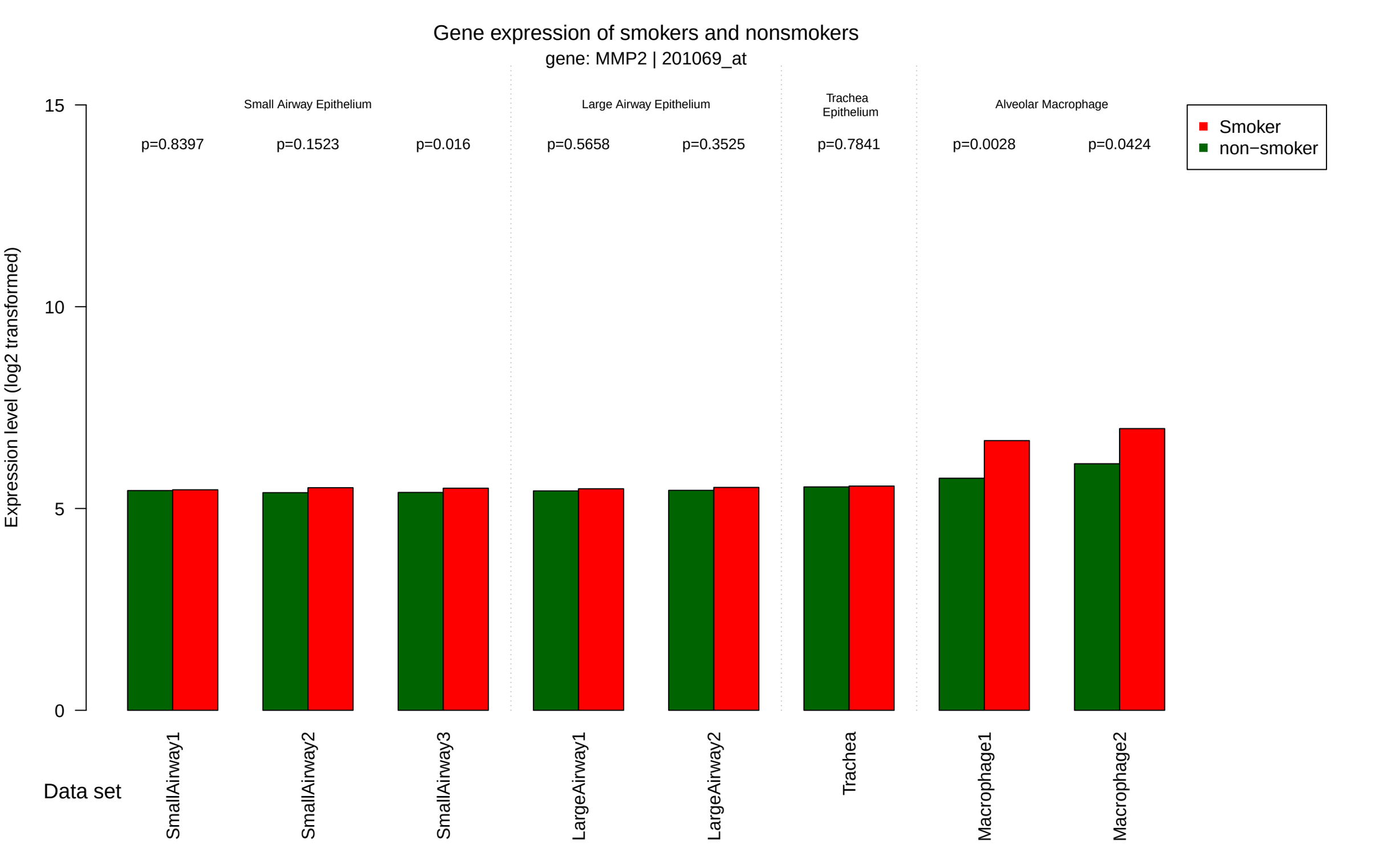
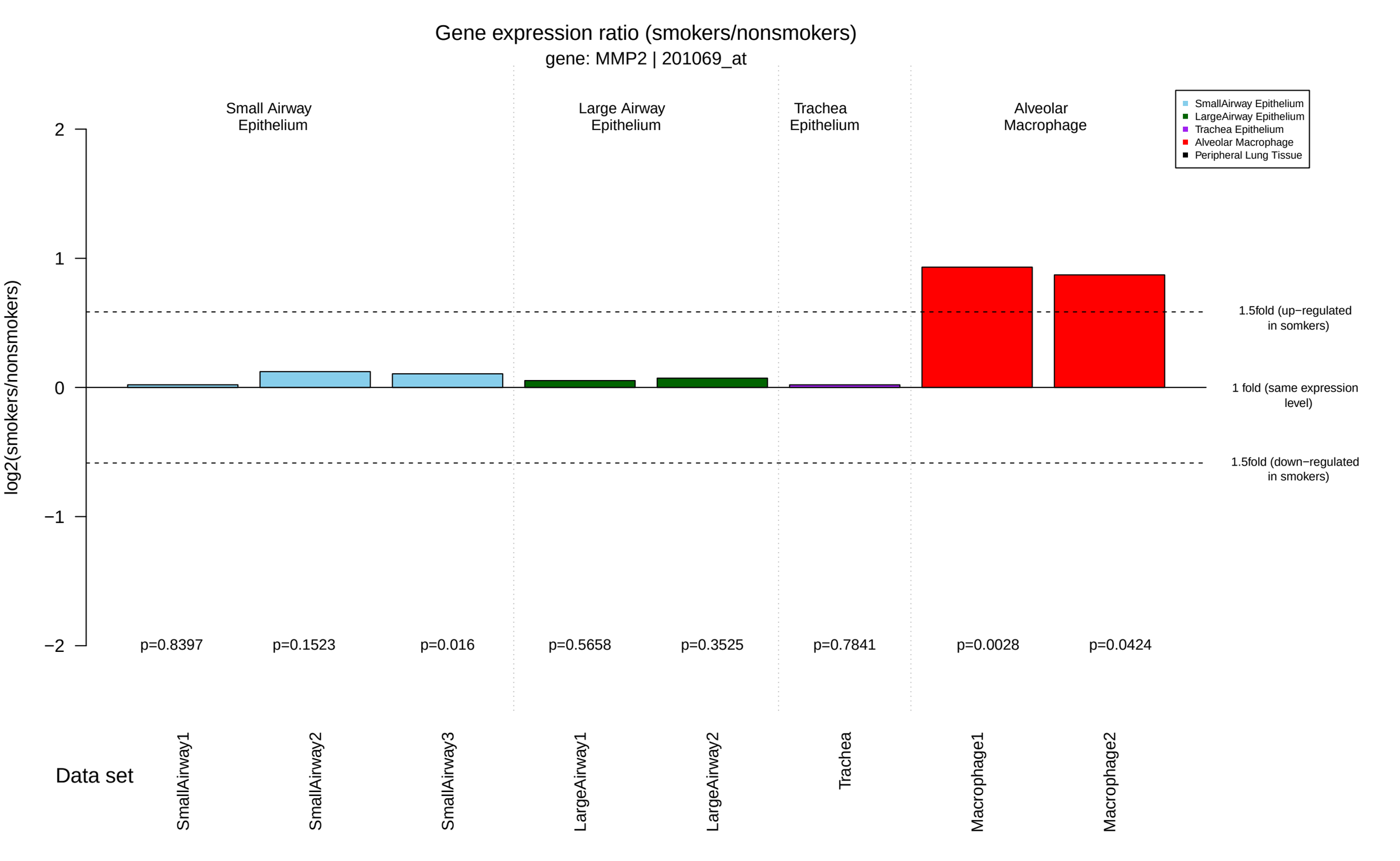
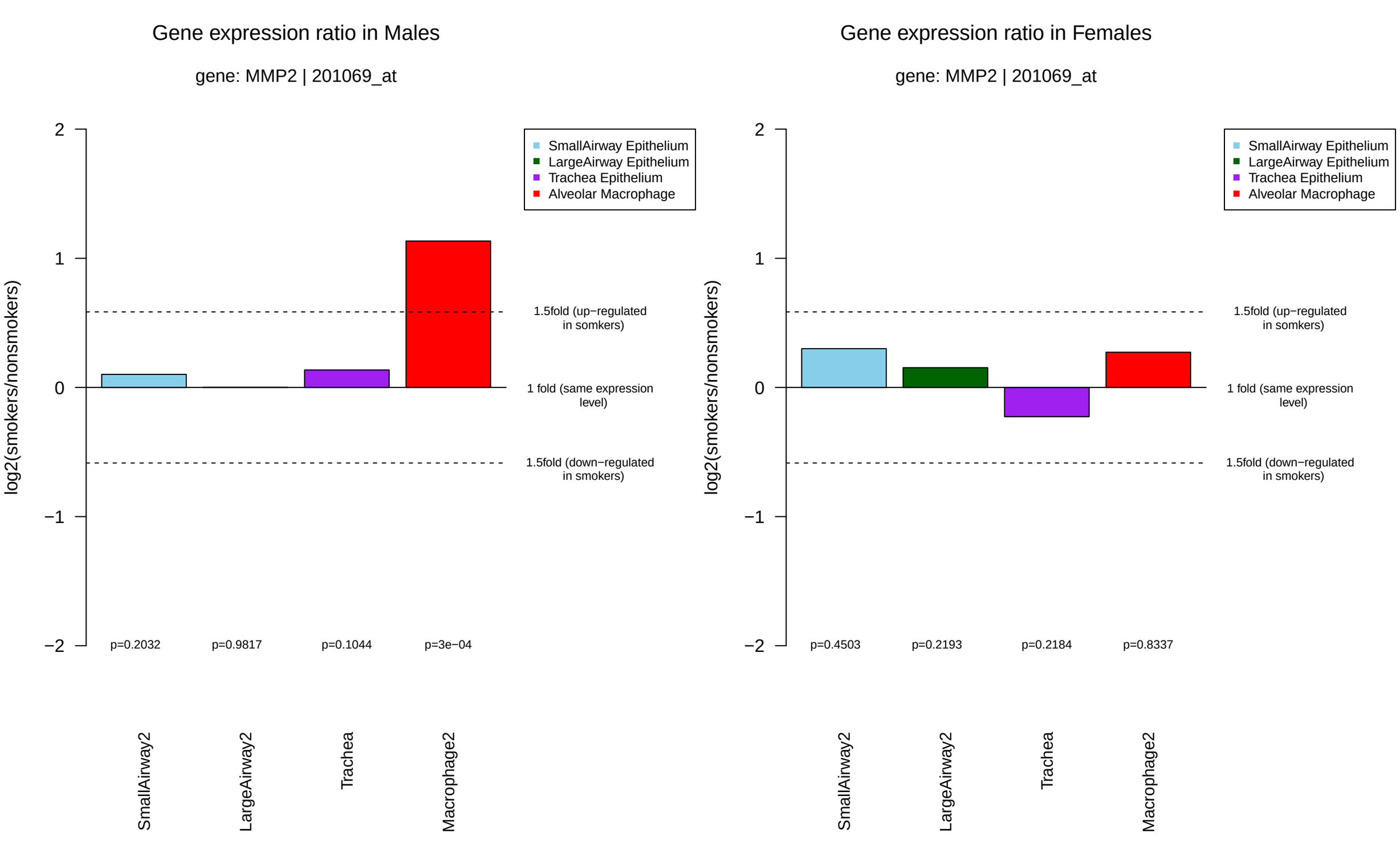
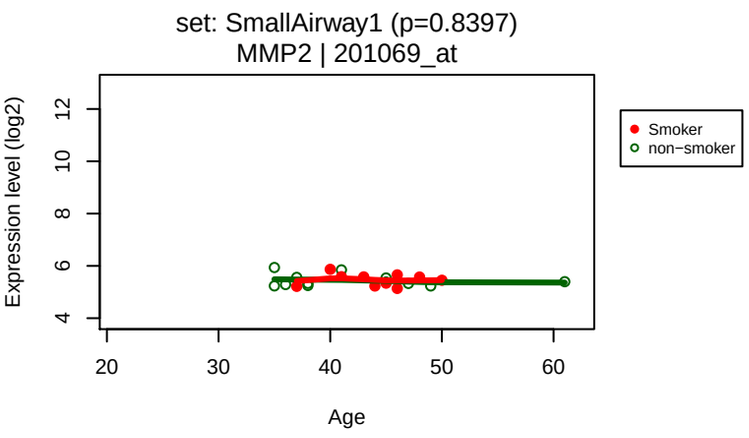
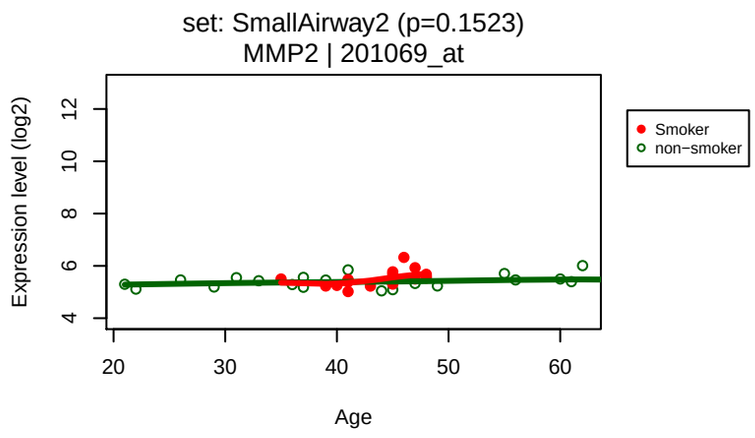
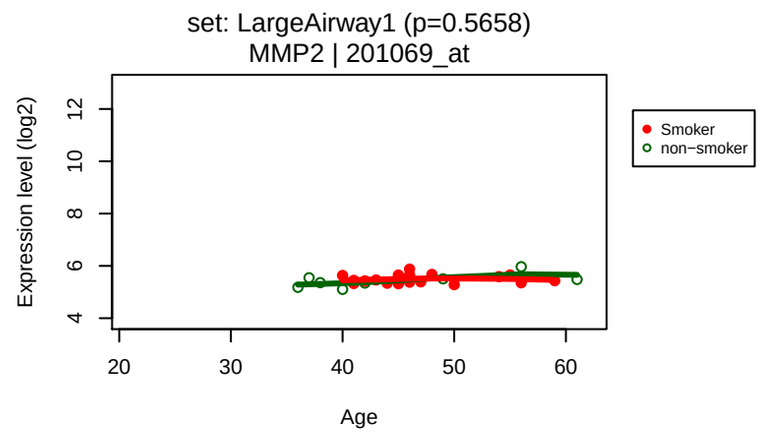

Gene expression of smokers and nonsmokers
gene: MMP2 | 201069_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8397
p=0.1523
p=0.016
p=0.5658
p=0.3525
p=0.7841
p=0.0028
p=0.0424
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

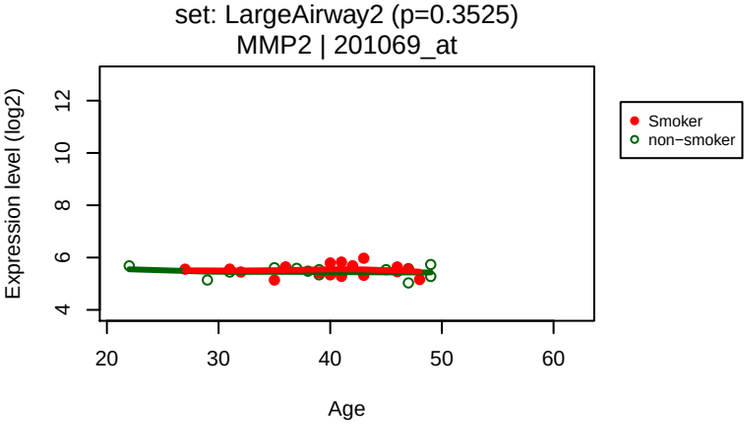

Gene expression ratio (smokers/nonsmokers)
gene: MMP2 | 201069_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8397
p=0.1523
p=0.016
p=0.5658
p=0.3525
p=0.7841
p=0.0028
p=0.0424
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

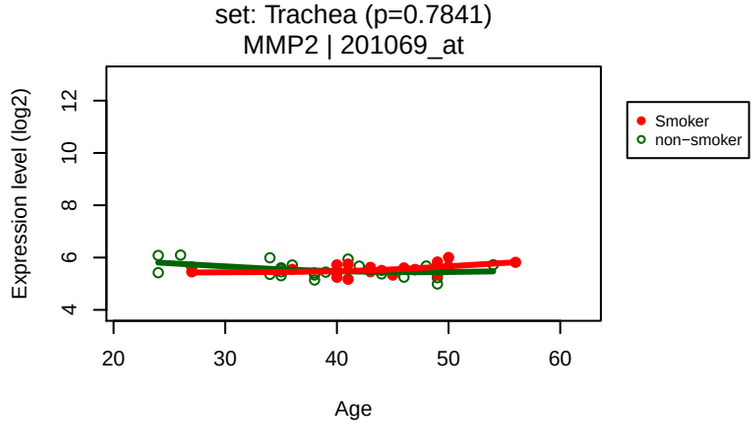

Gene expression ratio in Males
gene: MMP2 | 201069_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2032
p=0.9817
p=0.1044
p=3e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP2 | 201069_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4503
p=0.2193
p=0.2184
p=0.8337
SmallAirway2
LargeAirway2
Trachea
Macrophage2

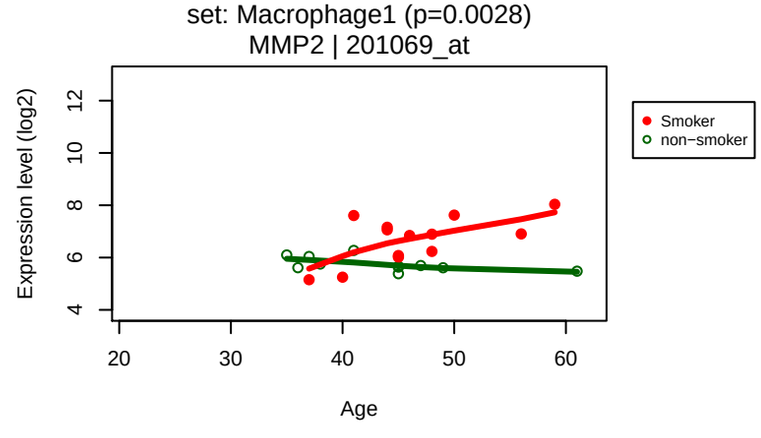

set: SmallAirway1 (p=0.8397)
MMP2 | 201069_at
Expression level (log2)
Age
Smoker
non−smoker

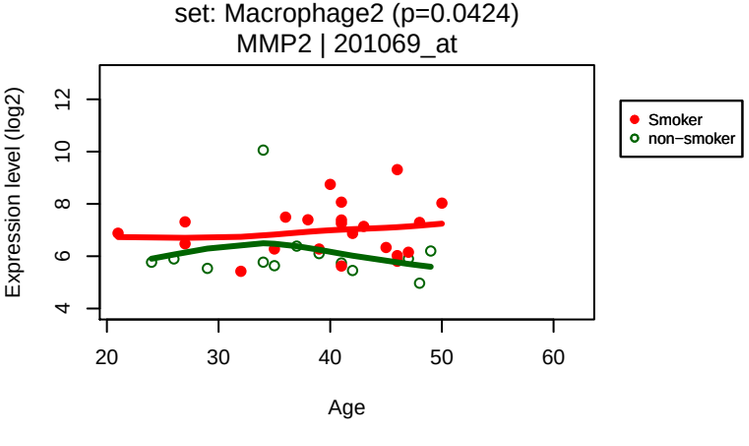

set: SmallAirway2 (p=0.1523)
MMP2 | 201069_at
Expression level (log2)
Age
Smoker
non−smoker

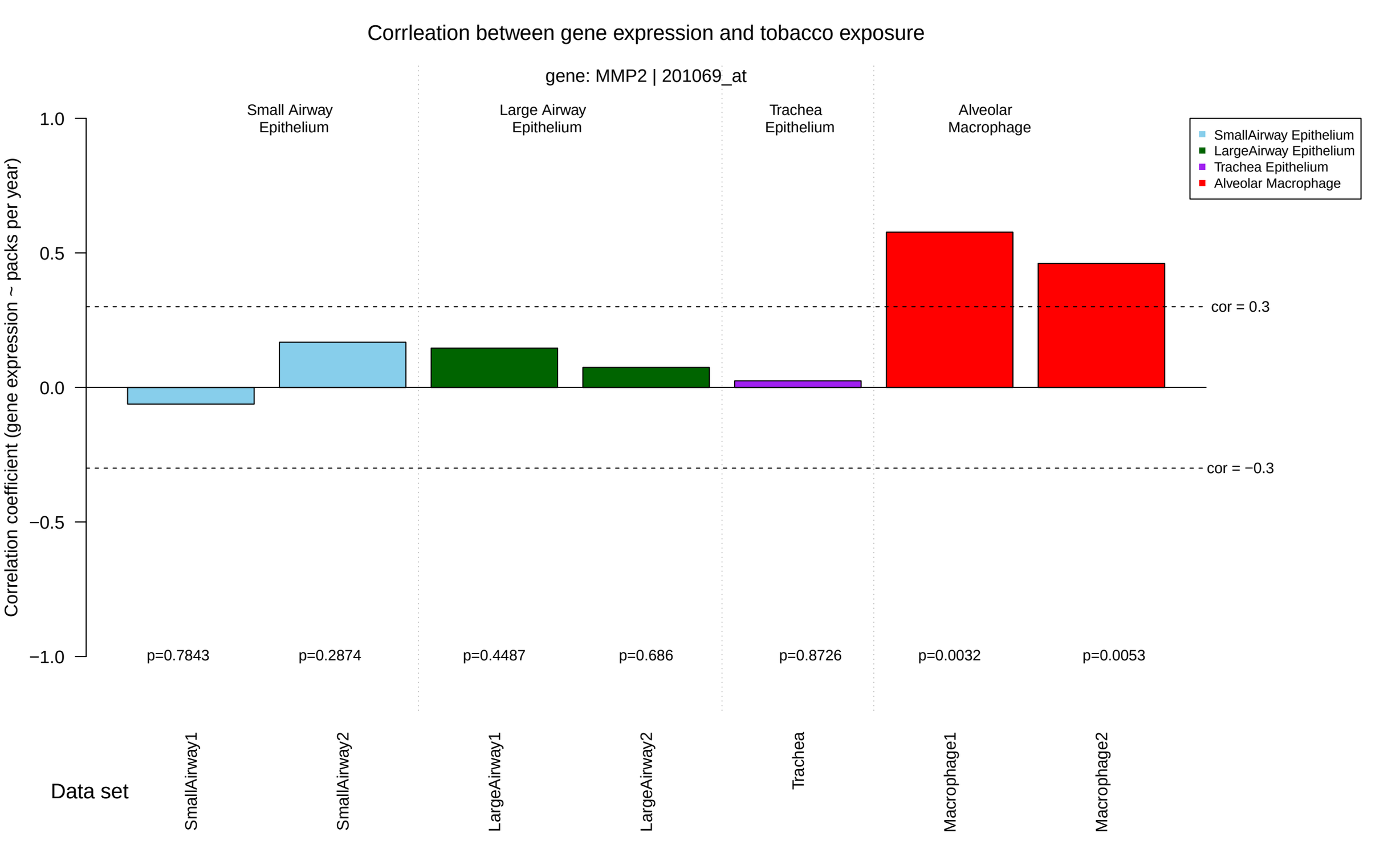

set: LargeAirway1 (p=0.5658)
MMP2 | 201069_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3525)
MMP2 | 201069_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7841)
MMP2 | 201069_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0028)
MMP2 | 201069_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0424)
MMP2 | 201069_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP2 | 201069_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7843
p=0.2874
p=0.4487
p=0.686
p=0.8726
p=0.0032
p=0.0053
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2